

BABU BANARASI DAS **UNIVERSITY**

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SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

IT Student and Project Tracking Using MongoDB

Submitted to:

Mr. Harendra Singh

Department: BCA (DS & AI)

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Submitted by:

ADNAN AHMAD

Roll no:

1250258048

ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher, Mr. Harendra Singh, for giving me the opportunity to work on this NoSQL project titled

"IT Student & Project Tracker".

This project helped me understand the fundamental concepts of NoSQL databases and learn how MongoDB is utilized to create, retrieve, update, sort, and delete data. I also learned how various query operators such as greater than, less than, equal to, AND, and OR are practically applied in MongoDB.

I am highly thankful to my teacher for the invaluable guidance and continuous support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, an IT Student and Project Tracker has been created using MongoDB. The database contains information about 50 students, including their roll number, name, course, skills, completed projects, and city.

Different MongoDB operations have been performed on the student data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte have also been demonstrated.

OBJECTIVES

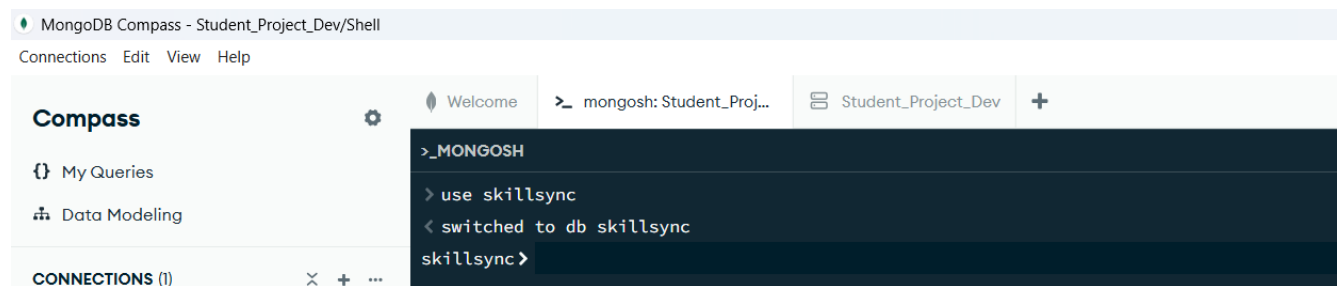
- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

1. CREATE DATABASE

MongoDB Query:

use skillsync

Explanation: Creates or switches to the skillsync database.

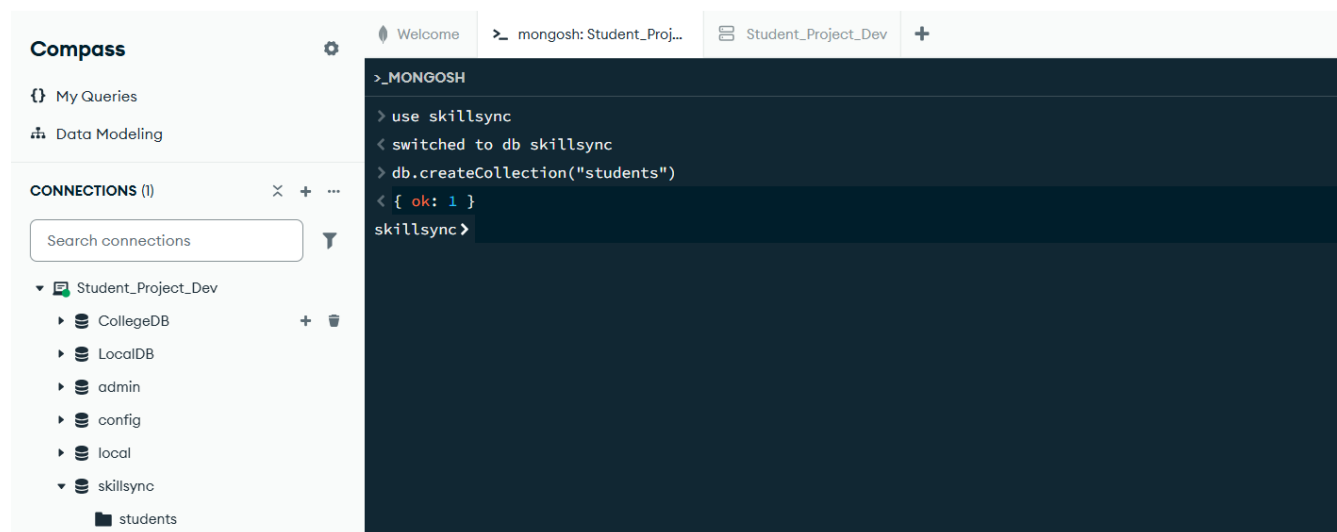


2. CREATE COLLECTION

MongoDB Query:

db.createCollection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

```
db.students.insert Many([ ])
```

```
{roll: 1, name: "Anamika", course: "BCA DS AI", skills: ["Python", "Pandas"],  
projects_completed: 3, city: "Lucknow"},
```

```
{roll: 2, name: "Kabir", course: "BCA DS AI", skills: ["Node.js", "Express"],  
projects_completed: 2, city: "Delhi"},
```

```
{roll: 3, name: "Rohan", course: "BCA DS AI", skills: ["Java", "SQL"], projects_completed: 5,  
city: "Pune"},
```

```
{roll: 4, name: "Priya", course: "BCA DS AI", skills: ["Scala", "MongoDB"],  
projects_completed: 1, city: "Lucknow"},
```

```
{roll: 5, name: "Amit", course: "BCA", skills: ["HTML", "CSS", "JavaScript"],  
projects_completed: 4, city: "Noida"},
```

```
{roll: 6, name: "Sneha", course: "BCA DS AI", skills: ["Python", "Machine Learning"],  
projects_completed: 3, city: "Bangalore"},
```

```
{roll: 7, name: "Rahul", course: "BCA", skills: ["C++", "Data Structures"], projects_completed:  
2, city: "Mumbai"},
```

```
{roll: 8, name: "Neha", course: "BCA DS AI", skills: ["Python", "SQL", "Tableau"],  
projects_completed: 6, city: "Delhi"},
```

```
{roll: 9, name: "Vikram", course: "MCA", skills: ["Java", "Spring Boot"], projects_completed: 3,  
city: "Hyderabad"},
```

```
{roll: 10, name: "Aditi", course: "BCA DS AI", skills: ["Node.js", "MongoDB", "React"],  
projects_completed: 5, city: "Gurgaon"},
```

```
{roll: 11, name: "Karan", course: "BCA", skills: ["PHP", "MySQL"], projects_completed: 1, city:  
"Kanpur"},
```

```
{roll: 12, name: "Meera", course: "BCA DS AI", skills: ["Python", "Seaborn", "Pandas"],  
projects_completed: 4, city: "Lucknow"},
```

```
{roll: 13, name: "Dev", course: "MCA", skills: ["AWS", "Docker", "Linux"], projects_completed:  
2, city: "Pune"},
```

```
{roll: 14, name: "Isha", course: "BCA DS AI", skills: ["JavaScript", "Express", "MongoDB"],  
projects_completed: 3, city: "Delhi"},
```

{roll: 15, name: "Arjun", course: "BCA", skills: ["Java", "Android Studio"], projects_completed: 4, city: "Bangalore"}.

{roll: 16, name: "Tanya", course: "BCA DS AI", skills: ["Python", "TensorFlow"], projects_completed: 2, city: "Mumbai"}.

{roll: 17, name: "Sahil", course: "MCA", skills: ["C#", ".NET"], projects_completed: 5, city: "Noida"}.

{roll: 18, name: "Pooja", course: "BCA DS AI", skills: ["R", "SQL", "Excel"], projects_completed: 3, city: "Lucknow"}.

{roll: 19, name: "Gaurav", course: "BCA", skills: ["C", "C++"], projects_completed: 1, city: "Jaipur"}.

{roll: 20, name: "Kriti", course: "BCA DS AI", skills: ["Python", "Flask", "SQL"], projects_completed: 4, city: "Delhi"}.

{roll: 21, name: "Nitin", course: "MCA", skills: ["Go", "Kubernetes"], projects_completed: 3, city: "Hyderabad"}.

{roll: 22, name: "Riya", course: "BCA DS AI", skills: ["Scala", "Spark"], projects_completed: 2, city: "Pune"}.

{roll: 23, name: "Manish", course: "BCA", skills: ["HTML", "Bootstrap"], projects_completed: 1, city: "Kanpur"}.

{roll: 24, name: "Sanya", course: "BCA DS AI", skills: ["Python", "Django"], projects_completed: 5, city: "Bangalore"}.

{roll: 25, name: "Aakash", course: "MCA", skills: ["Node.js", "GraphQL"], projects_completed: 4, city: "Gurgaon"}.

{roll: 26, name: "Simran", course: "BCA DS AI", skills: ["Java", "Hibernate"], projects_completed: 2, city: "Mumbai"}.

{roll: 27, name: "Yash", course: "BCA", skills: ["JavaScript", "Vue.js"], projects_completed: 3, city: "Lucknow"}.

{roll: 28, name: "Shruti", course: "BCA DS AI", skills: ["Python", "NLP"], projects_completed: 6, city: "Delhi"}.

{roll: 29, name: "Tushar", course: "MCA", skills: ["Ruby on Rails", "PostgreSQL"], projects_completed: 2, city: "Pune"}.

{roll: 30, name: "Nidhi", course: "BCA DS AI", skills: ["MongoDB", "Express", "Angular", "Node.js"], projects_completed: 5, city: "Hyderabad"}.

{roll: 31, name: "Harsh", course: "BCA", skills: ["C++", "Qt"], projects_completed: 1, city: "Noida"}.

{roll: 32, name: "Mansi", course: "BCA DS AI", skills: ["Python", "PowerBI"], projects_completed: 4, city: "Lucknow"}.

{roll: 33, name: "Prateek", course: "MCA", skills: ["Java", "Kotlin"], projects_completed: 3, city: "Bangalore"}.

{roll: 34, name: "Aarti", course: "BCA DS AI", skills: ["Data Science", "NumPy"], projects_completed: 2, city: "Delhi"}.

{roll: 35, name: "Varun", course: "BCA", skills: ["PHP", "Laravel"], projects_completed: 5, city: "Mumbai"}.

{roll: 36, name: "Kajal", course: "BCA DS AI", skills: ["Python", "Scikit-Learn"], projects_completed: 3, city: "Pune"}.

{roll: 37, name: "Saurabh", course: "MCA", skills: ["Cybersecurity", "Python"], projects_completed: 4, city: "Gurgaon"}.

{roll: 38, name: "Divya", course: "BCA DS AI", skills: ["SQL", "NoSQL", "MongoDB"], projects_completed: 2, city: "Jaipur"}.

{roll: 39, name: "Ayush", course: "BCA", skills: ["Swift", "iOS"], projects_completed: 1, city: "Kanpur"}.

{roll: 40, name: "Rakhi", course: "BCA DS AI", skills: ["Python", "FastAPI"], projects_completed: 5, city: "Lucknow"}.

{roll: 41, name: "Chirag", course: "MCA", skills: ["Rust", "WebAssembly"], projects_completed: 3, city: "Hyderabad"}.

{roll: 42, name: "Adnan", course: "BCA DS AI", skills: ["Python", "Data Analysis", "MongoDB"], projects_completed: 4, city: "Lucknow"}.

{roll: 43, name: "Shweta", course: "BCA", skills: ["UI/UX", "Figma"], projects_completed: 2, city: "Delhi"}.

{roll: 44, name: "Tarun", course: "BCA DS AI", skills: ["Hadoop", "Spark", "Scala"], projects_completed: 6, city: "Bangalore"}.

{roll: 45, name: "Roshni", course: "MCA", skills: ["Azure", "Python"], projects_completed: 3, city: "Pune"}.

{roll: 46, name: "Deepak", course: "BCA DS AI", skills: ["React Native", "Firebase"], projects_completed: 4, city: "Mumbai"}.

{roll: 47, name: "Pallavi", course: "BCA", skills: ["HTML", "Tailwind CSS"],
projects_completed: 1, city: "Noida"}},

{roll: 48, name: "Gautam", course: "BCA DS AI", skills: ["C++", "Blockchain"],
projects_completed: 2, city: "Gurgaon"}},

{roll: 49, name: "Anjali", course: "MCA", skills: ["Java", "Microservices"], projects_completed:
5, city: "Hyderabad"}},

{roll: 50, name: "Rishabh", course: "BCA DS AI", skills: ["Node.js", "Socket.io"],
projects_completed: 3, city: "Lucknow"}},

_MONGOSH

```
db.Students.insertMany([
  {"roll": 1, "name": "Anamika", "course": "BCA DS AI", "skills": ["Python", "Pandas"], "projects_completed": 3, "city": "Lucknow"},
  {"roll": 2, "name": "Kabir", "course": "BCA DS AI", "skills": ["Node.js", "Express"], "projects_completed": 2, "city": "Delhi"},
  {"roll": 3, "name": "Rohan", "course": "BCA DS AI", "skills": ["Java", "SQL"], "projects_completed": 5, "city": "Pune"},
  {"roll": 4, "name": "Priya", "course": "BCA DS AI", "skills": ["Scala", "MongoDB"], "projects_completed": 1, "city": "Lucknow"},
  {"roll": 5, "name": "Amit", "course": "BCA", "skills": ["HTML", "CSS", "JavaScript"], "projects_completed": 4, "city": "Noida"},
  {"roll": 6, "name": "Sneha", "course": "BCA DS AI", "skills": ["Python", "Machine Learning"], "projects_completed": 3, "city": "Bangalore"},
  {"roll": 7, "name": "Rahul", "course": "BCA", "skills": ["C++", "Data Structures"], "projects_completed": 2, "city": "Mumbai"},
  {"roll": 8, "name": "Neha", "course": "BCA DS AI", "skills": ["Python", "SQL", "Tableau"], "projects_completed": 6, "city": "Delhi"},
  {"roll": 9, "name": "Vikram", "course": "MCA", "skills": ["Java", "Spring Boot"], "projects_completed": 3, "city": "Hyderabad"},
  {"roll": 10, "name": "Aditi", "course": "BCA DS AI", "skills": ["Node.js", "MongoDB", "React"], "projects_completed": 5, "city": "Gurgaon"}])
```

Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

db.students.find()

Explanation: Retrieves documents from the students collection.

```
> db.students.find()
< {
  _id: ObjectId('6ab6997151fcd5651bc20ce8'),
  roll: 1,
  name: 'Anamika',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Pandas'
  ],
  projects_completed: 3,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20ce9'),
  roll: 2,
  name: 'Kabir',
  course: 'BCA DS AI',
  skills: [
    'Node.js',
    'Express'
  ],
  projects_completed: 2,
  city: 'Delhi'
}
```

5. EQUAL TO — \$eq

MongoDB Query:

```
db.students.find({city: {$eq: "Lucknow"}})
```

Explanation: Finds students who live exactly in "Lucknow".

```
> db.students.find({marks: {$eq: 88}})
< {
  _id: ObjectId('6aae9ea78f0f98eb52df221c'),
  roll: 1,
  name: 'Anamika',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aae9ea78f0f98eb52df2236'),
  roll: 27,
  name: 'Ananya',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({projects_completed: {$gt: 4}})
```

Explanation: Finds students who have completed strictly more than 4 projects.

```
>_MONGOSH
> db.students.find({projects_completed: {$gt: 4}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20cea'),
  roll: 3,
  name: 'Rohan',
  course: 'BCA DS AI',
  skills: [
    'Java',
    'SQL'
  ],
  projects_completed: 5,
  city: 'Pune'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20cef'),
  roll: 8,
  name: 'Neha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'SQL',
    'Tableau'
  ],
  projects_completed: 6,
  city: 'Delhi'
}
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.students.find({projects_completed: {$lt: 2}})
```

Explanation: Finds students with fewer than 2 completed projects.

```
>_MONGOSH
}
> db.students.find({projects_completed: {$lt: 2}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20ceb'),
  roll: 4,
  name: 'Priya',
  course: 'BCA DS AI',
  skills: [
    'Scala',
    'MongoDB'
  ],
  projects_completed: 1,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20cf2'),
  roll: 11,
  name: 'Karan',
  course: 'BCA',
  skills: [
    'PHP',
    'MySQL'
  ],
  projects_completed: 1,
  city: 'Kanpur'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.students.find({projects_completed: {$gte: 5}})
```

Explanation: Finds students who have completed 5 or more projects.

```
>_MONGOSH
    city: 'Noida'
  }
> db.students.find({projects_completed: {$gte: 5}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20cea'),
  roll: 3,
  name: 'Rohan',
  course: 'BCA DS AI',
  skills: [
    'Java',
    'SQL'
  ],
  projects_completed: 5,
  city: 'Pune'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20cef'),
  roll: 8,
  name: 'Neha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'SQL',
    'Tableau'
  ],
  projects_completed: 6,
  city: 'Delhi'
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

db.students.find({projects_completed: {\$lte: 3}})

Explanation: Finds students with 3 or fewer completed projects.

```
>_MONGOSH
> db.students.find({projects_completed: {$lte: 3}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20ce8'),
  roll: 1,
  name: 'Anamika',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Pandas'
  ],
  projects_completed: 3,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20ce9'),
  roll: 2,
  name: 'Kabir',
  course: 'BCA DS AI',
  skills: [
    'Node.js',
    'Express'
  ],
  projects_completed: 2,
  city: 'Delhi'
}
```

10. SEARCH WITH AN ARRAY

MongoDB Query:

```
db.students.find({skills: "Python"})
```

Explanation: Finds students who possess "Python" in their skills array.

```
>_MONGOSH
> db.students.find({skills: "Python"})
< {
  _id: ObjectId('6ab6997151fcd5651bc20ce8'),
  roll: 1,
  name: 'Anamika',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Pandas'
  ],
  projects_completed: 3,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20ced'),
  roll: 6,
  name: 'Sneha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Machine Learning'
  ],
  projects_completed: 3,
  city: 'Bangalore'
}
```

11. SEARCH USING IN OPERATOR

MongoDB Query:

```
db.students.find({skills: {$in: ["Java", "C++"]}})
```

Explanation: Finds students who possess either "Java" or "C++" in their skills array.

```
> _MONGOSH
> db.students.find({skills: {$in: ["Java", "C++"]}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20cea'),
  roll: 3,
  name: 'Rohan',
  course: 'BCA DS AI',
  skills: [
    'Java',
    'SQL'
  ],
  projects_completed: 5,
  city: 'Pune'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20cee'),
  roll: 7,
  name: 'Rahul',
  course: 'BCA',
  skills: [
    'C++',
    'Data Structures'
  ],
  projects_completed: 2,
  city: 'Mumbai'
}
```

12. SEARCH USING ALL (\$ALL) OPERATOR

MongoDB Query:

```
db.students.find({skills: {$all: ["Node.js", "Express"]}})
```

Explanation: Finds students who possess both "Node.js" and "Express" in their skills array simultaneously.

```
>_MONGOSH
> db.students.find({skills: {$all: ["Node.js", "Express"]}})
< {
  _id: ObjectId('6ab6997151fcd5651bc20ce9'),
  roll: 2,
  name: 'Kabir',
  course: 'BCA DS AI',
  skills: [
    'Node.js',
    'Express'
  ],
  projects_completed: 2,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20d05'),
  roll: 30,
  name: 'Nidhi',
  course: 'BCA DS AI',
  skills: [
    'MongoDB',
    'Express',
    'Angular',
    'Node.js'
  ],
  projects_completed: 5,
  city: 'Hyderabad'
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.students.find({$and: [{course: "BCA DS AI"}, {projects_completed: {$gt: 3}}]})
```

Explanation: Finds students enrolled in "BCA DS AI" who have also completed more than 3 projects.

```
>_MONGOSH
```

```
> db.students.find({$and: [{course: "BCA DS AI"}, {projects_completed: {$gt: 3}}]})
```

```
< {
  _id: ObjectId('6ab6997151fcd5651bc20cea'),
  roll: 3,
  name: 'Rohan',
  course: 'BCA DS AI',
  skills: [
    'Java',
    'SQL'
  ],
  projects_completed: 5,
  city: 'Pune'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20cef'),
  roll: 8,
  name: 'Neha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'SQL',
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.students.find({$or: [{city: "Pune"}, {city: "Bangalore"}]})
```

Explanation: Finds students who are either based in "Pune" or "Bangalore".

```
>_MONGOSH
> db.students.find({$or: [{city: "Pune"}, {city: "Bangalore"}]})
< {
  _id: ObjectId('6ab6997151fcd5651bc20cea'),
  roll: 3,
  name: 'Rohan',
  course: 'BCA DS AI',
  skills: [
    'Java',
    'SQL'
  ],
  projects_completed: 5,
  city: 'Pune'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20ced'),
  roll: 6,
  name: 'Sneha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Machine Learning'
  ],
  projects_completed: 3,
  city: 'Bangalore'
}
```

15. UPDATE RECORD USING SET(\$set) OPERATOR

MongoDB Query:

```
db.students.updateOne({roll: 1}, {$set: {city: "Noida"}})
```

Explanation: Updates the city of the student with roll number 1 to "Noida".

```
>_MONGOSH
}
> db.students.updateOne({roll: 1}, {$set: {city: "Noida"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

16. UPDATE RECORD USING INCREMENT(\$inc) OPERATOR

MongoDB Query:

```
db.students.updateOne({roll: 2}, {$inc: {projects_completed: 1}})
```

Explanation: Increases the project completion count by 1 for the student with roll number 2.

```
> db.students.updateOne({roll: 2}, {$inc: {projects_completed: 1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

17. UPDATE ARRAY USING PUSH(\$push) OPERATOR

MongoDB Query:

```
db.students.updateOne({roll: 3}, {$push: {skills: "React"}})
```

Explanation: Adds "React" to the skills array of the student with roll number 3.

```
>_MONGOSH
> db.students.updateOne({roll: 3}, {$push: {skills: "React"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

18. UPDATE ARRAY USING PULL(\$pull) OPERATOR

MongoDB Query:

```
db.students.updateOne({roll: 5}, {$pull: {skills: "HTML"}})
```

Explanation: Removes "HTML" from the skills array of the student with roll number 5.

```
> db.students.updateOne({roll: 5}, {$pull: {skills: "HTML"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

19. DELETE A SINGLE RECORD

MongoDB Query:

```
db.students.deleteOne({roll: 4})
```

Explanation: Deletes the record of the student whose roll number is 4.

```
> db.students.deleteOne({roll: 4})
< {
  acknowledged: true,
  deletedCount: 1
}
```

20. DELETE MULTIPLE RECORDS

MongoDB Query:

```
db.students.deleteMany({projects_completed: 0})
```

Explanation: Deletes all records of students who have completed exactly 0 projects.

```
> db.students.deleteMany({projects_completed: 0})
< {
  acknowledged: true,
  deletedCount: 0
}
```

21. SHORT DATA IN ASCENDING ORDER

MongoDB Query:

```
db.students.find().sort({name: 1})
```

Explanation: Sorts the list of all students alphabetically by their names from A to Z.

```
>_MONGOSH
> db.students.find().sort({name: 1})
< {
  _id: ObjectId('6ab6997151fcd5651bc20d00'),
  roll: 25,
  name: 'Aakash',
  course: 'MCA',
  skills: [
    'Node.js',
    'GraphQL'
  ],
  projects_completed: 4,
  city: 'Gurgaon'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20d09'),
  roll: 34,
  name: 'Aarti',
  course: 'BCA DS AI',
  skills: [
    'Data Science',
    'NumPy'
  ],
  projects_completed: 2,
  city: 'Delhi'
}
```

22. SHORT DATA IN DESCENDING ORDER

MongoDB Query:

```
db.students.find().sort({projects_completed: -1})
```

Explanation: Sorts all students from the highest number of projects completed to the lowest.

```
>_MONGOSH
> db.students.find().sort({projects_completed: -1})
< {
  _id: ObjectId('6ab6997151fcd5651bc20cef'),
  roll: 8,
  name: 'Neha',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'SQL',
    'Tableau'
  ],
  projects_completed: 6,
  city: 'Delhi'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20d13'),
  roll: 44,
  name: 'Tarun',
  course: 'BCA DS AI',
  skills: [
    'Hadoop',
    'Spark',
    'Scala'
  ],
  projects_completed: 6,
  city: 'Bangalore'
}
```

23. LIMIT QUERY RESULTS

MongoDB Query:

`db.students.find().limit(5)`

Explanation: Returns only the first 5 student records found in the collection.

```
>_MONGOSH
> db.students.find().limit(5)
< {
  _id: ObjectId('6ab6997151fcd5651bc20ce8'),
  roll: 1,
  name: 'Anamika',
  course: 'BCA DS AI',
  skills: [
    'Python',
    'Pandas'
  ],
  projects_completed: 3,
  city: 'Noida'
}
{
  _id: ObjectId('6ab6997151fcd5651bc20ce9'),
  roll: 2,
  name: 'Kabir',
  course: 'BCA DS AI',
  skills: [
    'Node.js',
    'Express'
  ],
  projects_completed: 3,
  city: 'Delhi'
}
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]
\$set	Replaces the value of a field with the specified value.	{\$set: {city: "Noida"}}
\$inc	Increments the value of the field by the specified amount.	{\$inc: {projects_completed: 1}}
\$push	Appends a specified value to an array.	{\$push: {skills: "React"}}
\$pull	Removes all array elements that match a specified query.	{\$pull: {skills: "HTML"}}

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. An IT Student and Project Tracker containing student technical profiles was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to sort data in Ascending & descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query IT student skills and project data efficiently.